AI workforce replacement & Employee attrition analysis

Key questions:

- How does AI usage level relate to productivity?
- Does increased AI adoption increase employee burnout?
- Which roles fear they are most at risk?
- Do employees with higher automation risk also have higher reskilling urgency?
- Can reskilling mitigate employee attrition and fear of replacement?

Summary

This project focuses on two different datasets. The first one is about employee burnout due to AI usage, and the second focuses on how likely AI is to replace roles in the industries.

Key findings in this project include:

- There is a dual effect in AI usage: increased number of AI tools is associated with higher productivity, and prolonged daily interaction with AI systems contributes to increased burnout
- Moderate AI usage may help maintain productivity without substantially increasing burnout, however, excessive daily reliance on AI assistance is associated with higher burnout levels
- Industries face the same level of challenge when it comes to fear of AI replacing their roles
- Industries should focus on reskilling employees to adapt to use AI in their work routines to offset the fear of replacement

Data Overview

The AI Worker Burnout & Attrition Risk dataset has 1,500 rows with 20 columns.

The AI Job Replacement & Skill Shift dataset has 15,000 rows with 20 columns.

Each of the rows from each of the datasets represents an employee with their productivity score, burnout score, how much they fear AI can replace their roles and so on.

Key variables used include:

- Productivity score
- Burnout score
- AI tools used per day
- Fear of AI replacement
- Remote work type
- Hours with AI assistance daily
- Industry
- Automation Risk Percent
- Reskilling Urgency Score
- Skill gap index

Data preparation

Python was the language of choice to carry out this project. Python libraries, such as pandas, matplotlib, and seaborn, were used to extract and work on the data as well as create visualizations.

Exploratory Data Analysis

- **How does AI usage level relate to productivity?**

The AI Burnout & Attrition Risk dataset contains employees with tech roles, such as data analysts, software engineers, cloud architects, and so on. The dataset includes countries such as India, Canada, Germany, USA, Singapore, Netherlands, UK, Australia, Brazil, and France. The distribution of countries is evenly spread.

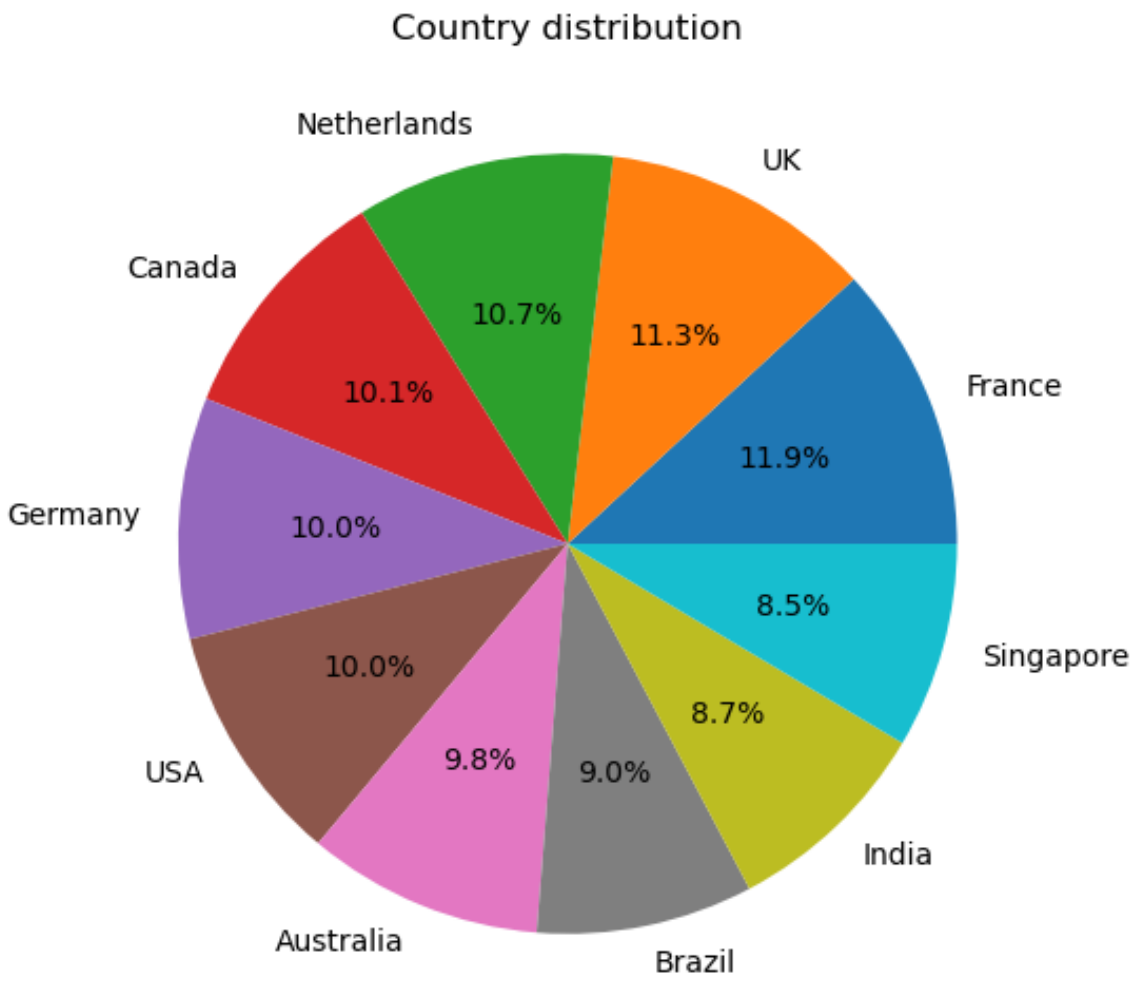

Fig. 1, countries' spread in the dataset

The top 3 most used AI tools include Midjourney (which is a popular generative AI program that creates images from natural language text prompts), Claude (which is a Large Language Model that is very good for writing, and complex reasoning, suitable for any tech role), and Github Copilot (which is an AI-powered pair programmer that helps write code faster). These tools can help developers and data analysts with their everyday tasks and also produce quality work. These tools also seem to create a fear of replacement due to how advanced they are. According to the dataset, employees experience some level of burnout when using AI tools. This is because employees must adapt to these tools and implement them in their routines. Employees have to be flexible, almost having to become a prompt engineer while they still have deadlines for their primary job. There is a constant learning curve when learning how to use just one tool.

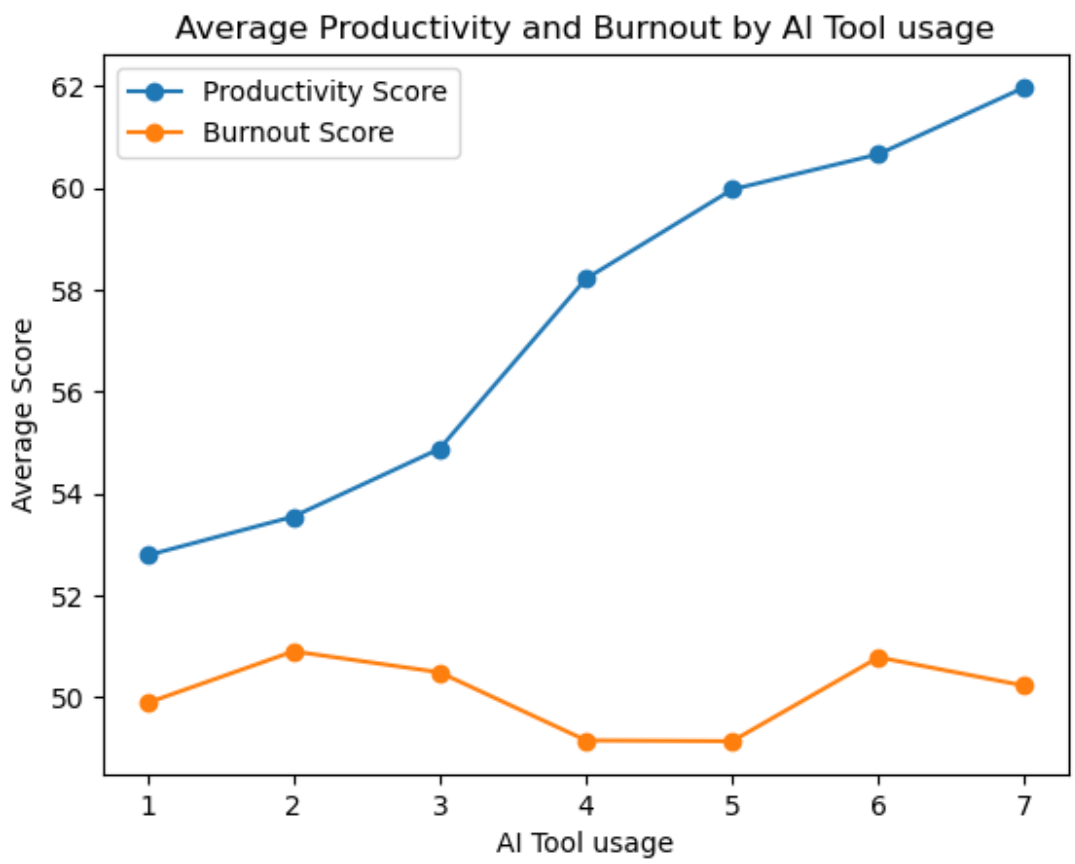

Fig. 2, productivity score and burnout score plot against AI tool usage

According to Fig. 2, productivity increases the more AI tools are used. At 7 tools, the productivity score is 62.

Fig. 2 also shows that burnout score increases from 1 to 2 AI tools used, but it drops when 3 or more AI tools are used. This suggests that there may be an optimal range of AI usage where productivity benefits are achieved without significantly increasing burnout, although further analysis indicates that time spent using AI plays a more critical role than the number of tools used. (Costa, 2024) In a survey by work management platform Asana, surveying 9,615 knowledge workers across Australia, France, Germany, Japan, the UK, and the US, it was found that "those that use six to 15 different apps in the workplace, 15% say they miss messages and notifications because of the number of tools". This finding refers to digital tools more broadly rather than exclusively AI tools, but it remains relevant as AI applications contribute to the overall increase in tool usage within modern workflows. This would reduce productivity overall because employees face greater pressure to avoid missing deadlines.

- **Does increased AI adoption increase employee burnout?**

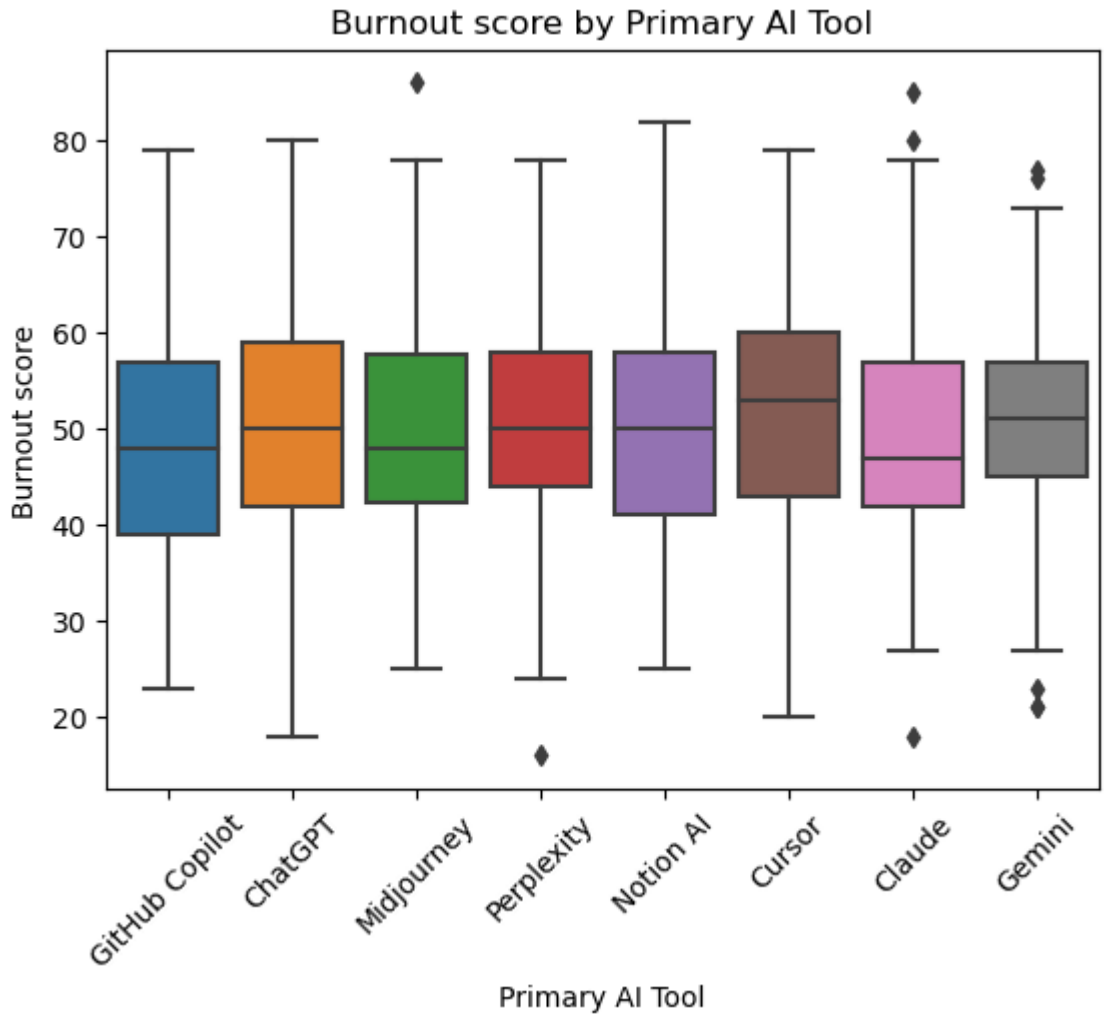

Fig. 3, box plot of burnout scores against the different types of AI

According to Fig. 2, if 6 or more tools are used, burnout scores will rise again. While AI can help automate repetitive tasks, the reality is rather different when it comes to how employees in certain industries use these. (Ranganathan and Ye, 2026) AI can "lead to cognitive fatigue, burnout, and weakened decision-making". Companies report that productivity has increased and that tasks have become easier to handle. However, there are areas where AI intensified employees' work. The first is task expansion, where employees are having to fill out other tasks that once belonged to other roles (for example, researchers are having to become more like engineers). In fact, the roles that employees feel most at risk are those that require research and credential checking of sources.

Secondly, multitask has increased, for example, while engineers write code, AI will generate an alternate version, and also run other tasks according to the backlog of tasks. It may sound efficient, but it requires constant attention to what task the AI is performing and testing the quality of those outputs.

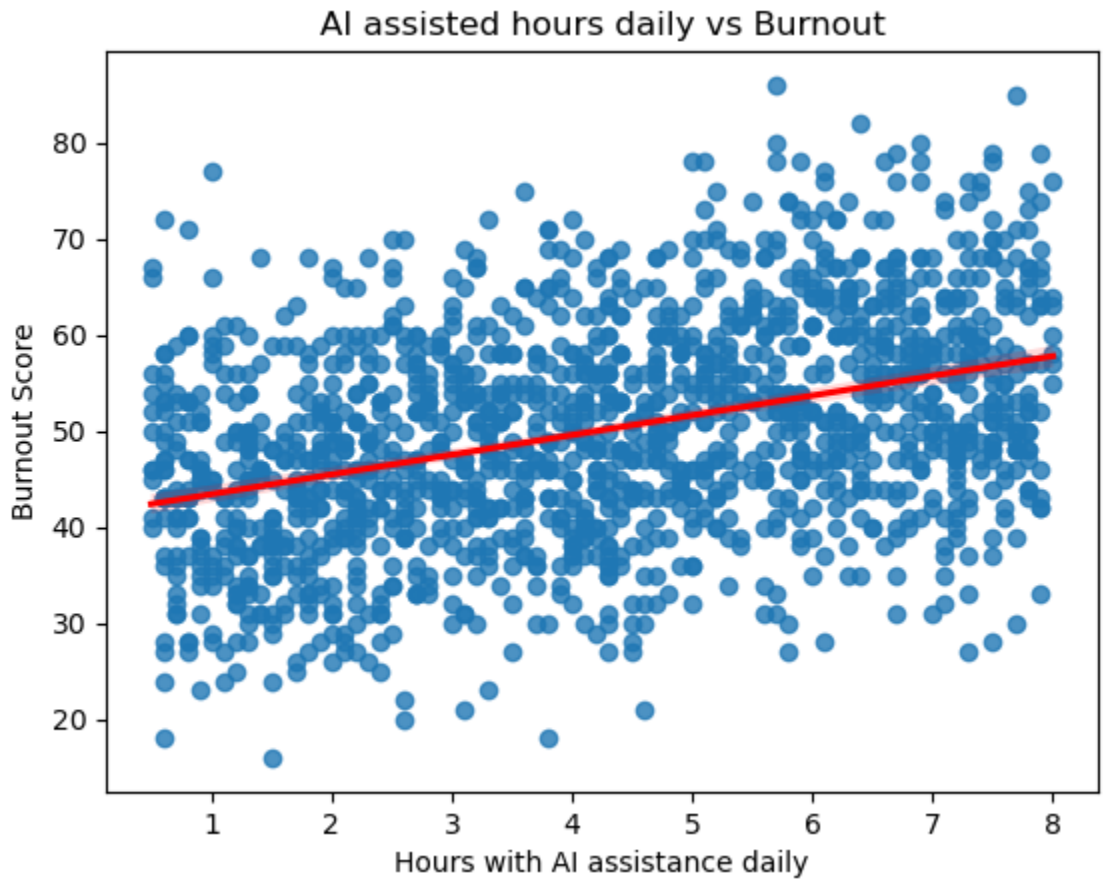

Fig. 4, scatterplot showing that there is a positive correlation between burnout scores and hours spent using AI at work

Fig. 4 is a scatterplot that shows how using AI for long periods of time can lead to higher burnout scores.

	Rank	job_role	burnout_score	fear_numeric
0	1	Backend Engineer	50.92	1.97
1	2	Software Engineer	51.27	1.93
2	3	ML Engineer	50.70	1.96
3	4	Data Scientist	50.55	1.93
4	5	AI Ethics Officer	49.71	1.95

Fig. 5, rank table showing which roles fear they are most at risk

Fig. 5 is a table generated by grouping all the roles in the AI Burnout & Attrition Risk dataset, and then a rank score was generated using burnout scores, and fear of AI replacement (which was first mapped into a numerical value).

Further research has been carried out using clustering. The objective is to find out the best compromise between how many AI tools should be used, and the hours that employees should spend at most. From this process, the key insight was that AI usage alone does not cause burnout, but rather long hours using AI cause higher burnout. However, using AI efficiently can lead to better productivity and higher job satisfaction.

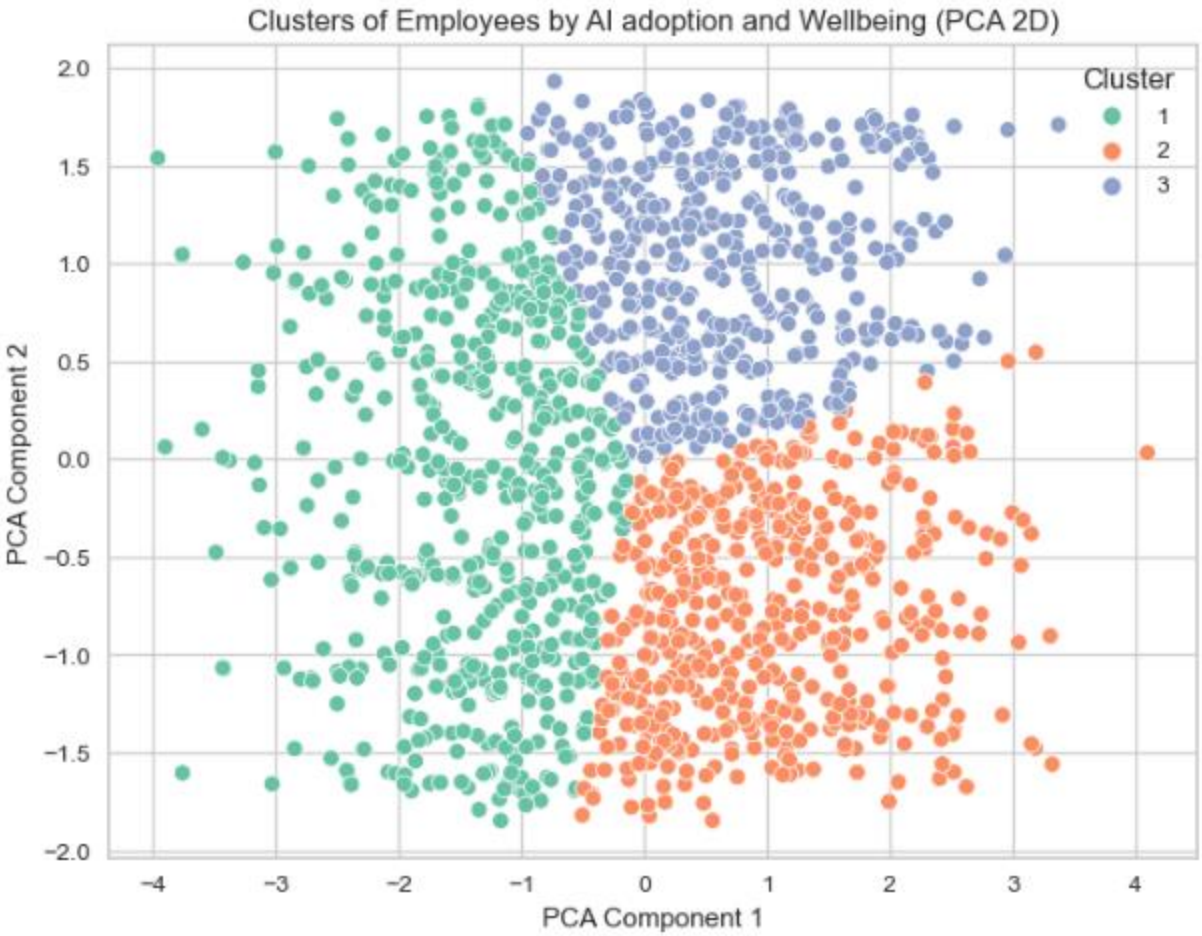

Fig. 6, Principal Component Analysis plot displaying how employees have been clustered in three categories

Fig. 6 shows the plot from the clustering process, and employees have been divided into 3 clusters.

cluster	ai_tools_used_per_day	hours_with_ai_assistance_daily	burnout_score	job_satisfaction_1_5
1	4.180851	5.702837	60.171986	2.800532
2	5.658436	3.494650	42.744856	3.769547
3	2.168889	3.190667	45.277778	3.534667

Fig. 7, table showing AI tools used, along with hours of AI assistance daily, burnout score, and job satisfaction on average

Fig. 7 explains the plot in Fig. 6 more clearly. The employees in the first cluster use 4 AI tools per day, spending approximately 6 hours a day, and the cluster yields 60% burnout score, with a job satisfaction average of 2.8/5, and 57.9% productivity score.
The employees in the second cluster use approximately 6 AI tools a day, spending 3.5

hours a day, and the cluster yields 42.7% burnout score, with a job satisfaction of 3.8/5, and a 60.6% productivity score.
The employees in the third cluster use 2 AI tools, spending 3 hours a day, and the cluster yields a 45% burnout score, with a job satisfaction of 3.5/5, and a 53.7% productivity score.

The employees in the second cluster are performing better than the other clusters, using an appropriate balance of AI tools and amount of time spent daily.

- **Which roles fear they are most at risk?**

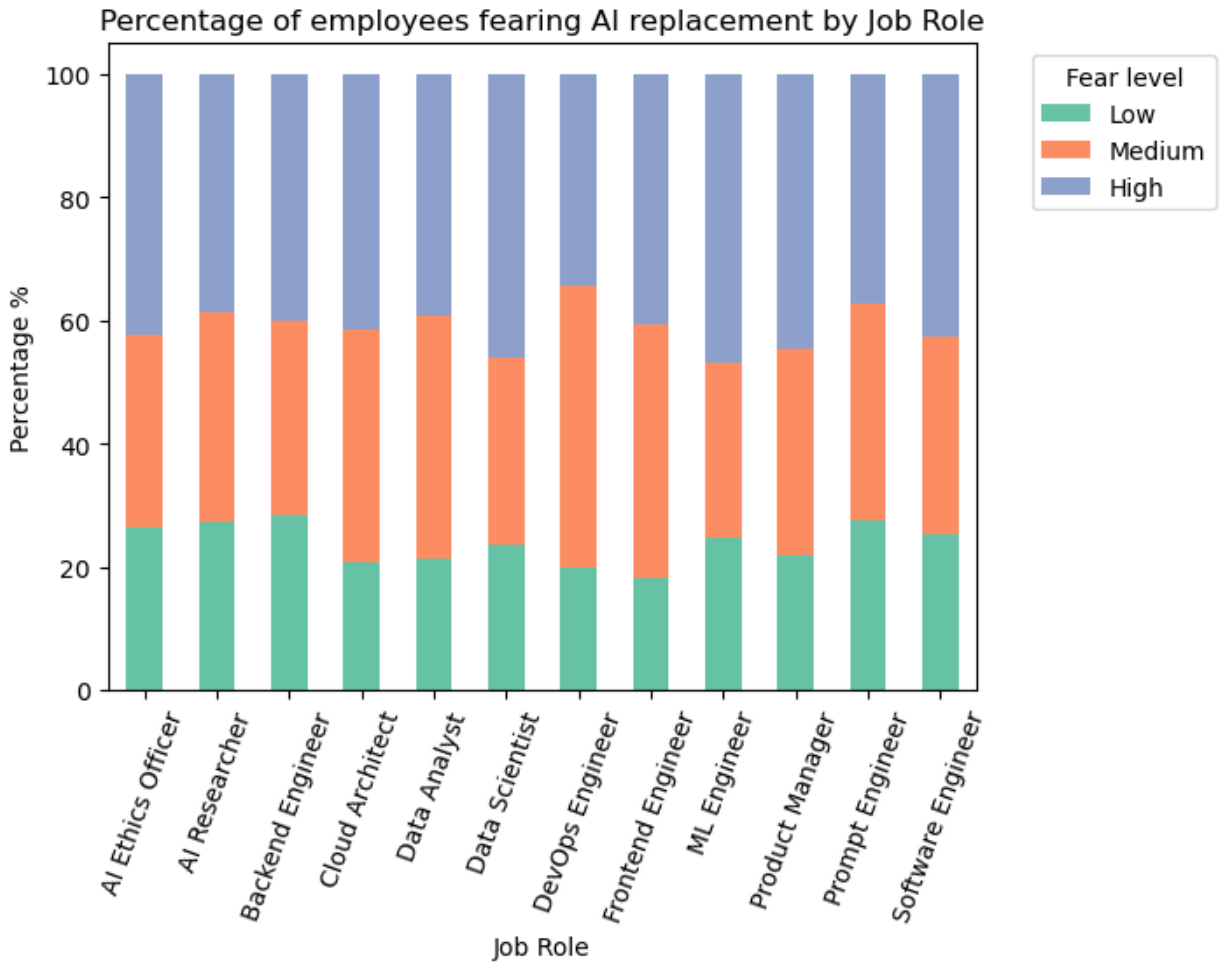

Fig. 8 stacked bar chart of job roles against percentage level of fear

Fig. 8 shows a stacked chart with the fear level across all the job roles found in that AI Burnout & Worker Attrition dataset. At first glance, data scientists, ML engineers, and product managers may show the biggest fear, but according to Fig. 5, software engineers are ranked with higher fear levels on average.

job_role	ai_tools_used_per_day	hours_with_ai_assistance_daily	fear_numeric
AI Ethics Officer	4.05	4.47	1.95
AI Researcher	3.98	4.13	1.93
Backend Engineer	4.08	4.30	1.97
Cloud Architect	3.95	4.35	1.83
Data Analyst	4.10	4.08	1.82
Data Scientist	4.21	3.96	1.93
DevOps Engineer	3.94	4.00	1.74
Frontend Engineer	4.05	4.12	1.77
ML Engineer	4.21	4.59	1.96
Product Manager	3.99	4.17	1.89
Prompt Engineer	4.30	4.52	1.92
Software Engineer	3.89	4.19	1.93

Fig. 9, table showing job roles' use of AI by how many tools and hours on average

According to Fig. 9, although AI ethics officers, and AI researchers use more AI tools and for longer periods of time on average, their fear level is very close to those of backend engineers. This suggests that higher exposure to AI does not necessarily translate into greater fear of replacement. Instead, employees who regularly interact with AI may develop a better understanding of its capabilities and limitations, leading to more balanced perceptions of risk.

- **Do employees with higher automation risk also have higher reskilling urgency?**

The AI Job Replacement & Skill Shift dataset contains other roles such as accounting, customer rep, and so on. These roles use AI differently and therefore may need different levels of reskilling depending on the role.

job_role	automation_risk_percent	reskilling_urgency_score
Truck Driver	60.740895	41.810564
Customer Support Rep	59.851052	41.595427
Marketing Specialist	45.438711	35.663488
Mechanical Engineer	45.397903	35.670092
Teacher	45.137897	35.722418
HR Manager	45.091339	35.443909
Accountant	44.979548	35.317330
Financial Analyst	44.704972	34.929181
Data Analyst	35.455807	31.608221
Software Engineer	34.853999	30.827850

Fig. 10, table showing that other non-tech roles face a higher need for reskilling

Fig. 10 shows that non-tech roles, such as truck drivers and marketing specialists, may need reskilling far more than tech roles, such as data analysts and software engineers.

Reskilling can refer to employee upskilling voluntarily, or the company may sponsor their employees to learn either in the office or at home.

Skill gap and AI replacement score also drive up reskilling urgency. Using ANOVA testing and post-hoc Tukey tests for each variable of skill gap index and AI replacement score confirm that the two are statistically significant on reskilling urgency score. There is a clear positive relationship between AI replacement score and reskilling score, and the same is for skill gap index. This means that employees recognize the perceived risk of AI replacing their tasks and the capability gaps that may need to be filled to stay ahead.

- **Can reskilling mitigate employee attrition and fear of replacement?**

The analysis suggests that AI-driven disruption contributes to increased employee uncertainty, which may elevate attrition risk. Roles experiencing higher levels of disruption and automation risk tend to report greater concern about job security and higher levels of burnout.

However, the findings also suggest that employees facing higher levels of automation risk and larger skill gaps demonstrate significantly greater urgency to reskill. While the datasets

do not directly measure whether reskilling reduces attrition, the strong relationship between risk, skill gap, and reskilling urgency suggests that reskilling may serve as a potential mitigating factor.

Overall, the results show that while AI adoption introduces challenges to workforce stability, proactive reskilling plays a critical role in enabling employees to navigate these changes.

Limitations

The datasets were from Kaggle, and the posters said that the data is synthetic. However, they also mentioned that the data is meant to simulate what is happening in real life.

Conclusion

This project explored the impact of AI usage on employee productivity, burnout, job security perceptions, and the role of reskilling to mitigate fear of replacement challenges. The findings reveal that AI adoption has a dual effect on employees. While increased use of AI systems is associated with higher productivity, prolonged daily interaction contributes to increased burnout. This suggests that the benefits of AI depend not only on adoption, but also on how it is integrated into daily workflows.

The analysis also highlights growing uncertainty around job security. This is more evident among roles that are very familiar with AI tools, as they realize that perhaps there will be a time when their role becomes redundant. That fear of replacement is different for every role.

The results also indicate that employees are not passive in response to that fear. There is a strong relationship between perceived risk, skill gaps, and reskilling urgency. Employees who actively develop new skills may be better equipped to manage technological change, potentially reducing their likelihood of leaving their roles.

Overall, the findings highlight that AI adoption presents both opportunities and challenges for the workforce. Organizations should focus not only on increasing AI adoption but also on supporting employees through effective reskilling initiatives and sustainable work practices to ensure long-term productivity and employee wellbeing.

Therefore, the most effective approach is not maximum AI adoption, but optimal integration, in terms of balancing tool usage and time spent to maximize productivity while

minimizing burnout. This highlights that the future work is not defined by AI alone, but by how organizations manage its implementation alongside human capability.

References

[1] Costa, M. (2024) *AI and workplace impact* [online]. BBC News. Available at: https://www.bbc.co.uk/news/articles/c93pz1dz2kxo (Accessed: 22 March 2026)

[2] Ranganathan, A. and Ye, X. M. (2026) *AI doesn't reduce work, it intensifies it* [online]. Harvard Business Review. Available at: https://hbr.org/2026/02/ai-doesnt-reduce-work-it-intensifies-it (Accessed: 23 March 2026)